

Question number	Answer	Additional guidance	Mark
7(a)(i)	An explanation that includes four of the following points: - pupil becomes smaller so less light enters the eye. (1) - circular muscles contract and radial muscles relax (1) - circular muscles arrange in concentric rings around pupil and radial muscles run radially (1) - pupil reflex is the reflex contraction and relaxation of the antagonistic muscles of the iris (1)		(3)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	A description that includes three of the following points: - rhodopsin absorbs light (1) - rhodopsin splits into opsin and retinal (1) sodium ions are taken up (1) - causing hyperpolarisation (1)		(4)

Levels-based mark schemes

Question number	Answer	Mark
7(b)	Indicative content Description of named responses • Many different photoresponses that affect the growth and development of plants. • Light triggering key events in a plants growth and development – photomorphogenesis. • Photoperiodisms – a physiological response to photoperiods eg seed germination. Chemical nature of responses Photoreceptor receive light, reacts to it and initiates a response. Phytochromes regulate many other responses to light eg seed germination. • Seeds need sunlight to grow. Far-red light makes plants think they are not getting sunlight. As a result, too much far-red on your seedlings may prevent germination altogether. • There are two forms of phytochrome. These are Pr and Pfr. These two pigments differ in their absorption peak. • Phytochrome red (Pr): It absorbs maximum light in the red region of the spectrum. It has wavelength of 660p Pr is the inactive form. • 2.Phytochrome far red (Pfr): It has the absorption peak in the far-red region at 730 p. Only Pfr phytochrome is physiologically active. • Metabolic activity and mitosis in embryos are stimulated by red light and inhibited by far-red light in light-sensitive seeds.	(6)

- plants also use light as a source of information, allowing for the adjustment of growth and development in response to changing environmental conditions. This process, termed photomorphogenesis, is regulated by a suite of photoreceptors

Detailed mechanism of responses

- Photoreceptor receive light, reacts to it and initiates a response
- Phytochrome red (Pr): It absorbs maximum light in the red region of the spectrum. It has wavelength of 660p Pr is the inactive form.
Phytochrome far red (Pfr): It has the absorption peak in the far-red region at 730 p. Only Pfr phytochrome is physiologically active.
- Phytochrome has two identical subunits. The chromophore of the phytochrome is photoreversible between the two isomers depending on the wavelength of light.
- Subcellular circadian rhythms include cycles in gene regulation (at the levels of transcription, transcript abundance, translation, and post-translational modification).
- Germination stimulated by the hormone gibberellin.
- Phytochrome red stimulates transcription.
- Leads to production of enzymes that metabolise active forms of gibberellins.

Level 1: Description of (a minimum of 2 specific) results from experimental data. eg. more exposure to far red light at all red light times reduces percentage germination.
Less exposure to far red light increases percentage of germination in general.

Level 2: Reference to specific chemical nature of the effect of red light and far red light on germination.

Level 3: Detailed mechanism for at least 1 photo-response at a sub-cellular level eg. Effect on transcription.

